

## SKILLS

## PROGRAMMING

Proficient:

MATLAB • QGIS • Google Earth Engine

Experienced:

Python • R • Stata • Tableau •  $\LaTeX$ 

## RESEARCH

Climate &amp; Disaster Risk Assessment

Earthquake Risk &amp; Loss Modelling

Deep Probabilistic Modelling

Weakly-Supervised Learning

Compositional Data Analysis

Geospatial Data Science

Graph Deep Learning

Spatiotemporal Analysis

## OPEN-SOURCE PROJECTS

- **VBCI** [📄★14] [📄66 cites] [📄242]  
Joint inference of multi-hazards & building damage from satellite images by causal dependency modelling (*Nature Communications*)
- **METEOR 2.5D: 72-GB Data** [📄960]  
Spatiotemporal maps of physical vulnerability in Least Developed Countries, 1975-2030 (AAAI'26)
- **DeepC4** [📄2k]  
Deep conditional census-constrained clustering for multi-task spatial disaggregation of urban morphology (AGU'24, ISPRS P&RS, *in review*)

## EDUCATION

## UNIVERSITY OF CAMBRIDGE

PH.D. ARTIFICIAL INTELLIGENCE FOR

THE STUDY OF ENVIRONMENTAL RISKS

M.RES. ENVIRONMENTAL DATA SCIENCE

Oct. 2022 - Sep. 2026 | Cambridge, UK

UKRI EPSRC CDT Studentship (\$290k)

**Dissertation:** Understanding Regional Dynamic Physical Risks, Advanced Geospatiotemporal Modelling of Exposure & Vulnerability Using Deep Constrained Clustering & Probabilistic Temporal Graph Learning with EO Data

## STANFORD UNIVERSITY

M.A. PUBLIC POLICY AND

M.S. CIVIL &amp; ENVIRONMENTAL ENG'G

EXECUTIVE EDUCATION CERTIFICATE

IN ENTREPRENEURSHIP &amp; INNOVATION

Sep. 2019 - Jun. 2022 | Stanford, CA, USA

Knight-Hennessy Scholar (\$240k) (69/4,424)

## UNIVERSITY OF THE PHILIPPINES

B.S. CIVIL ENGINEERING

Jun. 2013 - Jan. 2018 | Los Baños, PH

*magna cum laude*, college valedictorian (1/341)  
completed one-semester early

## TOKYO METROPOLITAN UNIVERSITY

EXCHANGE PROGRAM CERTIFICATE,

INTEGRATED RIVER ENGINEERING WORK

Oct. - Nov. 2017 | Tokyo, JP

## EXPERIENCE

## GERMAN AEROSPACE CENTER (DLR) | VISITING RESEARCHER

Jul. - Sep. 2024 | Apr. - Jun. 2026 | Munich, Germany | 6 months

- Led development of a geospatial ML framework & benchmark dataset for building vulnerability & change detection using satellite embeddings & census-based constraints.

## EARTHQUAKES &amp; MEGACITIES INITIATIVE (EMI) | URBAN RESILIENCE FELLOW

Mar. - Aug. 2022 | Quezon City, Philippines | 5 months

- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population for a scenario "Big One" M7.2 earthquake.

## STANFORD LAND, BUILDINGS, &amp; REAL ESTATE (LBRE) | SEISMIC RISK CONSULTANT

Mar. 2021 - Mar. 2022 | Stanford, CA, USA | 1 year

- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings at Stanford University under seismic scenarios up to magnitude 7.6, and developed a Tableau dashboard for stakeholder decision-making.

## FM GLOBAL, ENG'G &amp; RESEARCH | STRUCTURES &amp; NATURAL HAZARDS RESEARCHER

Jun. - Sep. 2021 | Norwood, MA, USA | 4 months

- Improved the firm's library of earthquake design hazard and maps by more than 246%, starting with 130 and ending with 320 maps (or 190 new maps) from over 130 countries.

## STANFORD UNIVERSITY, STRUCTURES AS SENSORS RESEARCH LAB |

GEOSPATIAL &amp; MACHINE LEARNING GRADUATE RESEARCHER

Jun. 2020 - Jun. 2021 | Stanford, CA, USA | 1 year

- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance landslide, liquefaction, & structural damage models across the USA.

## SOUTH BAY CONSULTING ENGINEERS | STRUCTURAL ENGINEERING INTERN

Jun. - Sep. 2020 | San Jose, CA, USA | 3 months

- Produced structural designs and resolved review comments while leading retrofit of a 9,000 ft<sup>2</sup> two-story home with a weak first story using a cantilevered column system.

## NEW STORY CHARITY, RESEARCH &amp; DEV'T TEAM | HOUSING RESEARCH FELLOW

Feb. - Jun. 2020 | San Francisco, CA, USA | 4 months

- Evaluated 7 affordable housing material innovations and led feasibility studies with 20 interdisciplinary fellows across finance, policy, architecture, engineering, and construction.

## ARUP | STRUCTURAL ENGINEER

Jul. 2018 - Jul. 2019 | Metro Manila, Philippines | 1 year

- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41-50 story high-rise developments.

## MAKATI DEVELOPMENT CORPORATION | PLANNING &amp; CONTROL ENG'G INTERN

May. - Jul. 2017 | Bonifacio Global City, Philippines | 3 months

- Built a VBA-Excel system tracking deliverables across 100+ concurrent projects.

## ENGINEERING CONGRESS | COMMITTEE HEAD

Jun. 2016 - Jun. 2017 | Los Baños, Philippines | 1 year

- Developed event invitation and registration strategies that increased participation fivefold (300 to 1,500 attendees) for major engineering seminars and quiz competitions.

## PUBLICATIONS

Nature Communications, AAAI AI for Social Impact, Environmental Research Letters,

ISPRS Photogrammetry & Remote Sensing (*in review*), Progress in Disaster Science (*in review*)

## SELECTED TALKS

Moody's Risk Management Solutions (RMS), Global Earthquake Model (GEM) Conference, RiskLayer, Univ. of the Phils. Los Baños (UPLB), LandAware Innovations Group WG5

## SCIENTIFIC REVIEWER

AAAI AI for Social Impact, ICLR ML for Remote Sensing, UNDRR-ISC Hazard Info Profiles,

Deep Learning Indaba, Cambridge J. of Sci. &amp; Policy, Stanford J. of Sci., Tech., &amp; Society